

MATAGAMI, QC, Canada

WMO: 718210

Lat: 49.7578N

Lon: 77.7933W

Elev: 922

StdP: 14.21

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.6	-24.1	-37.7	0.7	-29.1	-32.2	0.9	-23.6	25.1	2.2	22.5	1.3	4.4	250	0.707

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	83.7	66.3	80.2	64.6	76.9	62.9	69.7	78.8	67.7	75.6	65.7	73.1	10.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.8	101.9	73.2	64.8	95.0	71.7	62.8	88.3	69.8	34.4	79.2	32.7	75.7	31.1	73.0	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.9	19.4	17.5	DB	-37.3	90.4	5.3	3.0	-41.1	92.6	-44.2	94.4	-47.2	96.1	-51.0	98.2	
			WB	-37.4	73.5	5.0	2.2	-41.0	75.1	-43.9	76.4	-46.7	77.7	-50.4	79.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.3	0.0	2.5	14.8	30.2	46.2	57.4	61.9	59.7	51.9	39.5	24.7	9.3	
	DBStd	24.30	14.33	13.20	14.09	10.96	9.86	8.91	6.43	6.70	8.33	8.57	10.97	13.39	
	HDD50	7241	1551	1329	1092	598	194	30	1	4	74	347	760	1261	
	HDD65	11673	2016	1749	1556	1045	587	260	139	194	401	791	1209	1726	
	CDD50	1156	0	0	1	3	77	251	370	303	130	21	0	0	
	CDD65	114	0	0	0	0	4	31	43	28	8	0	0	0	
	CDH74	1390	0	0	0	2	112	436	475	276	87	2	0	0	
	CDH80	326	0	0	0	0	29	123	115	45	14	0	0	0	
Wind	WSAvg	8.3	8.2	8.5	8.9	9.0	8.6	8.0	7.6	7.5	7.9	8.6	8.8	8.2	
Precipitation	PrecAvg	33.7	1.9	1.4	1.7	2.2	2.9	3.2	4.4	3.3	4.0	3.1	2.7	2.1	
	PrecMax	40.6	6.3	2.7	3.1	4.8	5.6	6.2	6.6	5.5	7.6	5.9	4.4	5.2	
	PrecMin	22.7	0.5	0.4	0.7	0.6	0.6	1.2	2.5	0.6	2.1	0.8	0.6	0.3	
	PrecStd	5.1	1.2	0.7	0.7	1.1	1.2	1.2	1.3	1.3	1.3	1.2	0.9	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.3	41.6	53.8	65.5	84.1	89.0	88.8	84.4	81.5	70.8	54.2	39.5	
		MCWB	36.1	38.2	44.1	49.2	62.4	68.0	69.8	69.0	67.4	59.5	49.8	37.9	
	2%	DB	32.7	33.5	45.5	58.4	76.4	83.6	82.8	80.7	76.0	63.8	47.4	34.8	
		MCWB	32.0	31.5	38.1	45.3	57.8	65.2	66.6	66.5	65.0	56.7	44.7	33.9	
	5%	DB	28.0	29.1	40.1	51.8	70.9	79.1	79.2	76.9	71.3	58.2	42.3	32.5	
		MCWB	27.0	27.3	35.2	42.0	55.8	62.5	64.7	64.5	62.2	53.5	39.9	31.8	
	10%	DB	23.0	24.1	35.8	46.4	65.2	74.7	75.6	73.2	67.1	53.2	38.4	28.9	
		MCWB	22.0	22.5	31.6	39.2	52.6	60.8	63.2	62.7	60.2	49.6	36.6	28.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.5	37.6	48.7	52.7	66.3	71.7	72.4	71.6	69.9	61.7	51.2	38.3
			MCDB	37.2	39.6	50.4	59.4	78.3	83.8	83.5	80.5	78.4	67.1	53.2	39.0
2%		WB	32.0	31.4	39.4	46.3	60.9	68.4	69.5	68.9	66.7	57.7	45.0	33.9	
		MCDB	32.6	33.3	43.5	55.7	69.7	78.3	77.9	76.8	73.4	62.5	46.4	34.5	
5%		WB	26.9	27.2	35.5	43.0	57.7	65.7	67.3	66.6	63.9	54.1	40.4	31.8	
		MCDB	27.7	29.0	39.6	50.5	67.3	74.1	74.9	73.9	70.5	58.0	42.1	32.5	
10%		WB	21.9	22.5	32.1	39.6	54.5	63.0	65.2	64.3	60.4	49.9	36.6	28.0	
		MCDB	23.0	24.0	35.3	46.0	64.1	71.6	72.3	71.0	65.9	53.0	38.2	28.8	
Mean Daily Temperature Range	5% DB	MDBR	19.6	22.3	23.1	20.8	23.5	24.0	21.7	20.5	19.4	14.2	12.8	15.7	
		MCDBR	18.9	21.1	24.4	29.2	33.2	29.6	26.9	26.2	25.7	23.0	16.6	12.7	
		MCWBR	19.0	19.6	18.8	18.7	18.2	14.7	13.8	14.3	15.8	16.4	14.5	12.6	
	5% WB	MCDBR	19.0	20.7	21.3	25.8	27.2	24.6	22.2	21.9	22.6	20.7	15.4	12.1	
		MCWBR	19.2	19.5	17.4	18.0	16.9	14.0	13.1	13.2	15.3	15.9	14.1	12.1	
Clear-Sky Solar Irradiance	taub	0.276	0.288	0.300	0.360	0.351	0.370	0.398	0.377	0.369	0.322	0.319	0.256		
	taud	2.179	2.182	2.225	2.252	2.423	2.388	2.349	2.402	2.392	2.518	2.310	2.319		
	Ebn at Noon	239	269	288	280	288	282	271	272	262	259	222	237		
	Edh at Noon	24	31	36	39	35	36	37	34	31	23	22	18		
All-Sky Solar Radiation	RadAvg	389	733	1193	1491	1623	1811	1648	1368	1014	582	297	243		
	RadStd	62	95	108	129	168	133	163	87	91	70	34	43		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	+3.19	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page